SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

Duration: 1 Hour

QUESTIONS: 1 Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer :

1. Error Identification
Given:

- **Training MSE = 0.85**
- **Testing MSE = 5.20**

Since the testing error is much higher than the training error, the model is **overfitting**.

Possible Causes

- Overfitting due to excessive training.
- Model is too complex (too many layers/neurons).
- Training dataset is too small.
- Training and testing data have different distributions.
- Presence of noisy or irrelevant features.

2. Error Analysis & Reasoning
Analysis

- During training, the neural network learns the training data very well, giving a **low MSE (0.85)**.
- During testing, the model cannot generalize to unseen data, resulting in a **high MSE (5.20)**.
- This large difference indicates that the model has memorized the training data instead of learning general patterns.

Reason

- High model complexity.
- Insufficient training samples.
- Lack of regularization.
- Poor feature selection.

3. Correction Strategy

The following corrective measures can reduce the testing error:

1. **Use Regularization**
    - Apply **L1 or L2 regularization** to reduce overfitting.
2. **Apply Dropout**
    - Randomly deactivate neurons during training to improve generalization.
3. **Increase Training Data**
    - Use more training samples or perform data augmentation.
4. **Early Stopping**
    - Stop training when the validation error starts increasing.
5. **Reduce Model Complexity**
    - Use fewer hidden layers or neurons.
6. **Feature Selection**
    - Remove noisy and irrelevant features.
7. **Cross-Validation**
    - Validate the model using k-fold cross-validation for better generalization.

4. Interpretation of Results

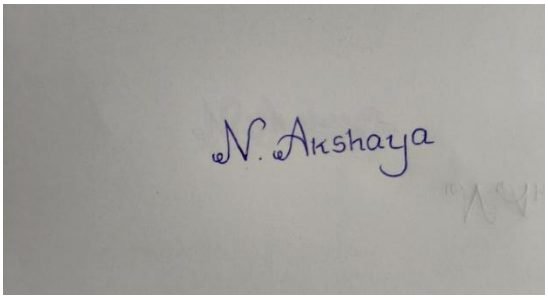
- The **training MSE (0.85)** indicates the model fits the training data well.
- The **testing MSE (5.20)** indicates poor performance on unseen data.
- The significant gap between training and testing errors confirms **overfitting**.
- Applying regularization, dropout, early stopping, increasing training data, and simplifying the model will improve generalization and reduce the testing MSE.

Conclusion:

- **Training MSE:** 0.85
- **Testing MSE:** 5.20
- **Error Identified:** Overfitting
- **Main Causes:** Complex model, limited training data, lack of regularization, noisy features.
- **Corrective Measures:** Regularization, Dropout, Early Stopping, More Data, Feature Selection, Cross-Validation.
- **Expected Outcome:** Reduced testing error and improved model generalization.

Code of Conduct:

I **N.AKSHAYA(192425129)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 – Exemplary (5)**	**Level 3 – Proficient (4)**	**Level 2 – Developing (2-3)**	**Level 1 – Beginning (0-1)**
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided